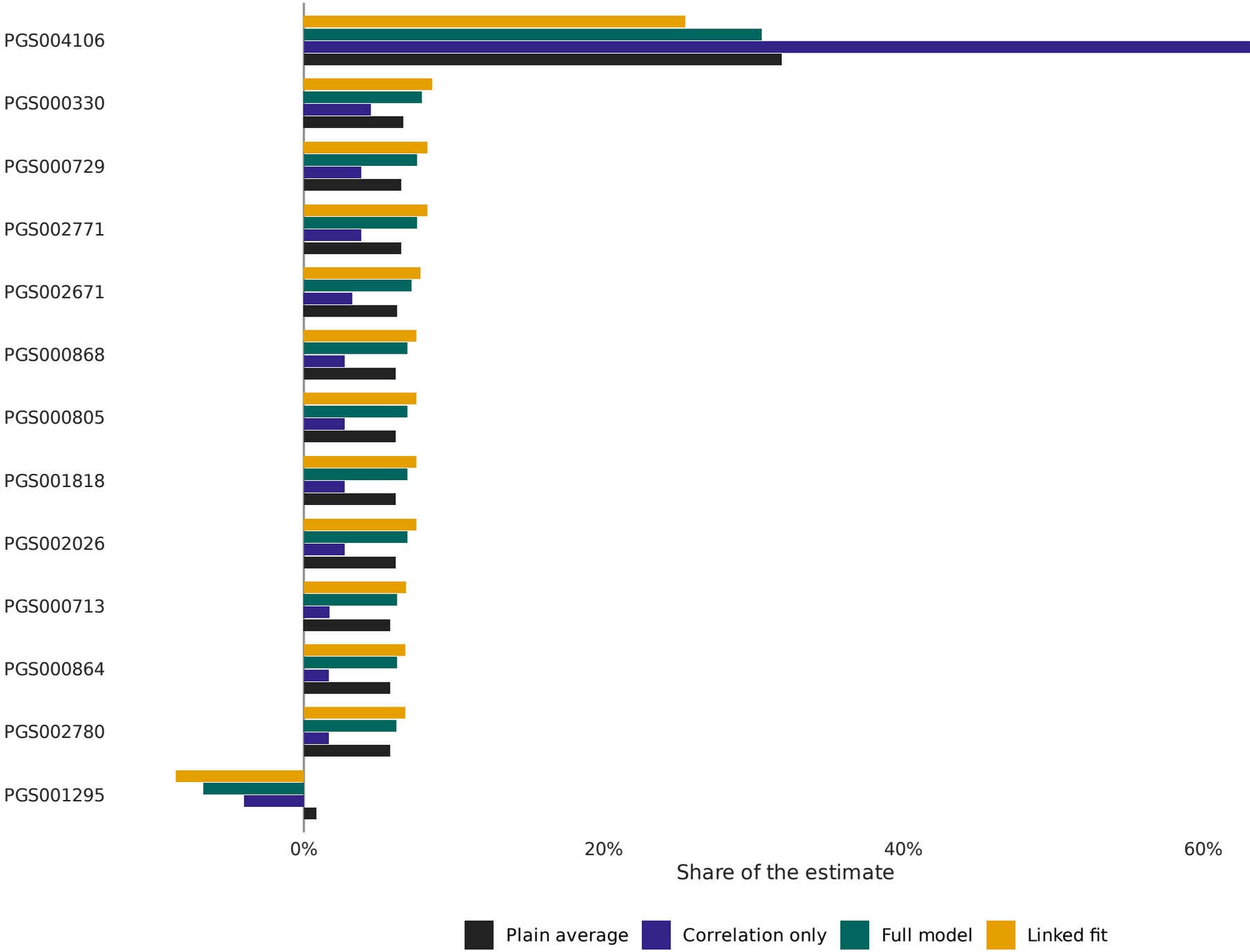
7. Why the estimate moves

Plain average 0.001. Correlation only (ρ = 0.6, σ² = 0) -0.010. Full model 0.004. Linked fit 0.006.



Shares are the row sums of the marginal precision matrix. They can be negative. With σ² = 0, PGS004106 does not share an exact label, so it takes most of the weight. PGS001295 shares the non-European label UK Biobank with the eleven and is pulled the other way. Estimating σ² brings PGS004106 back toward the plain average. The linked fit uses the chained groups.